# STUDENT CASE STUDY EXECUTION REPORT

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|---|---|
| **Department** | FED |
| **Subject** | DSA-2 |
| **Academic Year** | I-III Semester (2025-26) |
| **Faculty Name** | Ms. Mohammed Razia Alangir Banu |
| **Student Name** | Katta Yamini Sai |
| **Roll Number** | 2520030380 |
| **Branch** | CSE |
| **Course Outcome** | CO-4 |

**Case Study Topic: PetCare – Smart Veterinary Clinic & Pet Health Management System using Heap Sort, Greedy and Dynamic Programming**

**Work Done Summary:**

In this case study, advanced sorting and optimization techniques were implemented for the PetCare – Smart Veterinary Clinic & Pet Health Management System. Pet records, treatment plans, and healthcare services were organized and ranked using Heap Sort, Counting Sort, and Radix Sort based on treatment priority, health scores, and medical costs. Greedy Algorithm (Knapsack) was applied to select the most beneficial treatments within a pet owner's budget, while Dynamic Programming techniques such as 0/1 Knapsack, Longest Common Subsequence (LCS), and Matrix Chain Multiplication (MCM) were used to optimize treatment planning and healthcare recommendations. The implementation demonstrates efficient sorting, optimal decision-making, and healthcare optimization by considering time complexity, space usage, stability, and problem characteristics.

**Implementation Details:**

Created a dataset of pets containing treatment cost, health priority score, and medical history records.
Implemented Heap Sort to rank pets based on treatment priority and emergency levels.
Applied Counting Sort to efficiently sort pet health scores within a fixed range.
Implemented Radix Sort to organize treatment costs, billing records, and pet identification numbers.
Used the Greedy Knapsack Algorithm to select the maximum number of treatments within a pet owner's budget.
Implemented Longest Common Subsequence (LCS) to compare pet medical histories and recommend suitable treatment plans.
Applied Matrix Chain Multiplication (MCM) to optimize healthcare cost calculations and treatment scheduling.

10. Demonstrated efficient healthcare service coordination and emergency routing.

**RESULT : SCREENSHOTS/OUTPUT**

x

Sample Output

```
=== PETCARE MANAGEMENT SYSTEM ===

Health Priority Scores Before Heap Sort:
[85, 60, 95, 70, 80]

After Heap Sort:
[60, 70, 80, 85, 95]

Health Scores After Counting Sort:
[1, 2, 2, 3, 4, 4, 5]

Treatment Costs After Radix Sort:
[1200, 2200, 3500, 4500, 8000]
```

```
Selected Treatments:
Treatment Cost = ₹1000
Treatment Cost = ₹1500
Treatment Cost = ₹2000
Total Cost = ₹4500

LCS Length = 4
Minimum MCM Cost = 18000
```

**Github Link:https://github.com/yaminikatta18/DSA_CASE-STUDIES**

| **Student Signature** | **Faculty Signature** |
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| **Course Outcome** | CO-5 |

**Case Study Topic:PetCare – Smart Veterinary Clinic & Pet Health Management System using Advanced Web Development and Deployment**

**Work Done Summary:**

In this case study, the **PetCare – Smart Veterinary Clinic & Pet Health Management System** was designed and deployed as a web-based application by applying **CO-5 syllabus concepts of Advanced Web Development and Deployment**. The system provides digital support for **pet registration, appointment booking, doctor consultation details, vaccination reminders, treatment record management, and healthcare service access** through responsive web pages. Modern frontend concepts such as **semantic HTML, CSS styling, responsive layouts, JavaScript-based interaction, and user-friendly navigation** were used to improve usability. Deployment concepts were applied by preparing the project for hosting through **GitHub and online publishing platforms**. The implementation demonstrates effective use of web technologies, interface design, responsiveness, and deployment practices for veterinary clinic management.

**Implementation Details:**

1. Created a **PetCare web application interface** for managing pet registration, appointments, treatment records, and clinic services.
2. Designed responsive pages such as **Home, Pet Registration, Appointment Booking, Doctor Details, Vaccination Schedule, and Treatment History**.
3. Used **HTML and CSS** to build a structured, attractive, and user-friendly frontend for the veterinary clinic management system.
4. Implemented **JavaScript** for form validation, interactive navigation, and dynamic display of pet and appointment details.
5. Applied **responsive web design techniques** to ensure the PetCare system works properly on desktop, tablet, and mobile devices.
6. Organized **pet health records, doctor schedules, and service details** in a clear and accessible interface for pet owners and clinic staff.
7. Prepared the project for deployment by structuring files properly and publishing the PetCare application through **GitHub / online hosting platforms**.
8. Evaluated the system based on **usability, responsiveness, maintainability, and suitability** for smart veterinary healthcare management applications.
9.

**RESULT : SCREENSHOTS/OUTPUT**

## Sample Output

```
======================================
 PETCARE - SMART VETERINARY CLINIC SYSTEM
======================================
1. Register Pet
2. Display Pets
3. Add Doctor
4. Display Doctors
5. Book Appointment
6. Display Appointments
7. Add Vaccination Record
8. Display Vaccination Records
9. Add Treatment Record
10. Display Treatment Records
11. Generate Bill
12. Exit
Enter your choice: 1
```

```
Enter Pet ID: 101
Enter Pet Name: Bruno
Enter Owner Name: Yamini
Enter Pet Type: Dog
Enter Pet Age: 4
Pet registered successfully.

Enter your choice: 3
Enter Doctor ID: 201
Enter Doctor Name: Dr. Ravi
Enter Specialization: Surgery
Enter Available Time: 10AM - 2PM
Doctor added successfully.

Enter your choice: 5
Enter Appointment ID: 301
Enter Pet ID: 101
Enter Doctor ID: 201
Enter Appointment Date: 20-06-2026
```

```
Enter Reason for Visit: Vaccination
Appointment booked successfully.

Enter your choice: 7
Enter Pet ID: 101
Enter Vaccine Name: Anti-Rabies
Enter Due Date: 25-06-2026
Vaccination record added successfully.

Enter your choice: 9
Enter Pet ID: 101
Enter Treatment Name: General Checkup
Enter Treatment Cost: 500
Treatment record added successfully.

Enter your choice: 11
Enter Pet ID for bill generation: 101
```

```
Enter Pet ID: 101
Enter Treatment Name: General Checkup
Enter Treatment Cost: 500
Treatment record added successfully.


Enter your choice: 11
Enter Pet ID for bill generation: 101


--- BILL DETAILS ---
General Checkup - ₹500.0
Total Bill = ₹500.0
```

**GitHub link: https://github.com/yaminikatta18/DSA_CASE-STUDIES**

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| **Course Outcome** | CO-6 |

**Case Study Topic: PetCare – Smart Veterinary Clinic & Pet Health Management System using Non-Linear Data Structures**

**Summary:**

In this case study, graph-based algorithms and optimization techniques were implemented for the **PetCare – Smart Veterinary Clinic & Pet Health Management System**. The system was designed to model veterinary clinic operations such as pet-doctor connectivity, appointment scheduling, emergency service routing, treatment dependency mapping, and healthcare service optimization using graph representations. Algorithms such as **Breadth First Search (BFS), Depth First Search (DFS), Dijkstra's Algorithm, Bellman-Ford Algorithm, and Floyd-Warshall Algorithm** were applied to analyze clinic connectivity, find shortest treatment or service paths, optimize healthcare workflows, and improve emergency response planning. The implementation demonstrates efficient graph traversal, path optimization, service accessibility analysis, and network-based decision-making by considering time complexity, space usage, and suitability for veterinary healthcare applications.

# Implementation Details:

1. Created a graph-based dataset representing **pets, doctors, appointments, treatment services, and emergency healthcare connections** in the PetCare system.
2. Implemented **Breadth First Search (BFS)** to traverse clinic service connections and explore reachable healthcare services from a given pet or doctor node.
3. Applied **Depth First Search (DFS)** to analyze treatment dependency paths and connected healthcare modules within the clinic system.
4. Used **Dijkstra's Algorithm** to find the shortest path between veterinary services, such as selecting the quickest route for treatment planning or emergency handling.
5. Implemented **Bellman-Ford Algorithm** to compute shortest service paths in scenarios involving variable treatment costs or weighted healthcare connections.
6. Applied **Floyd-Warshall Algorithm** to determine all-pairs shortest paths among doctors, pets, and service units for complete clinic network optimization.
7. Analyzed service accessibility, emergency route planning, and healthcare workflow efficiency using graph traversal and shortest path techniques.

**RESULT : SCREENSHOTS/OUTPUT**

Output

```
=====================================
   PETCARE HEALTH RANGE QUERY SYSTEM
=====================================

Pet Health Scores:
85 72 90 60 78 95 68 88

Range Query:
Minimum Health Score from index 2 to 6 = 60

Updating Health Score...
Pet at index 3 health score changed from 60 to 55

After Update:
Minimum Health Score from index 2 to 6 = 55

=====================================
Time Complexity:
Segment Tree Construction : O(n)
Range Query               : O(log n)
Update Operation          : O(log n)
=====================================
```

```
Enter your choice: 4
Enter Treatment ID: 201
Enter Treatment Name: Vaccination
Enter Treatment Cost: 1500
Treatment record inserted into AVL Tree successfully.

Enter your choice: 6
Enter Source Node: 0
Enter Destination Node: 1
Connection added successfully.

Enter your choice: 6
Enter Source Node: 0
Enter Destination Node: 2
Connection added successfully.

Enter your choice: 7
Enter starting node for BFS: 0
BFS Traversal: 0 1 2

Enter your choice: 8
Enter starting node for DFS: 0
DFS Traversal: 0 1 2
```

Github Link:https://github.com/yaminikatta18/DSA_CASE-STUDIES

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